

Process Synchronization, Critical Section Problem & Semaphores

Subject: CS401 | Unit 2: Concurrency & Threads

1. Executive Overview & Syllabus Scope

Comprehensive handwritten & typed lecture notes covering Peterson's Solution, Counting & Binary Semaphores, Mutex Locks,

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 2: Process Synchronization & Concurrency Control
- 1. Critical Section Problem: Mutual Exclusion, Progress, Bounded Waiting conditions
- 2. Peterson's Algorithm: Software-based solution for two concurrent processes
- 3. Hardware Synchronization: Atomic TestAndSet and CompareAndSwap instructions
- 4. Semaphores: wait and signal atomic operations; Counting vs Binary semaphores
- 5. Classical Synchronization Problems: Producer-Consumer, Readers-Writers, Dining Philosophers
- 6. Deadlocks & Banker's Safety Algorithm step-by-step verification trace

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.